Supharada Jundok

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Education

National Cheng Kung University — Tainan, Taiwan
M.S. in Computer Science and Information Engineering — *Sep 2026 – Present*

National Dong Hwa University — Hualien, Taiwan
B.S. in Computer Science and Information Engineering (International Program) — *Sep 2022 – Jul 2026*

Coursework: Introductory ML, Big Data Analytics, Operating Systems, Intro to Generative AI, Project Management.

Skills

ML & Deep Learning: Few-Shot Learning, CNNs, ResNet50, EfficientNet, GANs, PyTorch, Episodic Training, Scikit-learn, XGBoost, LightGBM, Feature Engineering
Computer Vision: Image Processing, Image Classification, Feature Extraction, Segmentation, OpenCV
Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, SHAP, Jupyter
Remote Sensing & GIS: QGIS, GeoTIFF, CHM Processing, UAV RGB, Rasterio
Programming & Tools: Python, C, C++, SQL, Git, GitHub, VS Code, Linux
Cybersecurity: SIEM, IDS, Network Security, Threat Analysis

Research Experience

ICMN Lab, National Dong Hwa University — Hualien, Taiwan
Undergraduate Research Assistant – Few-Shot Learning — *Sep 2025 – Sep 2026*

- Built a custom CAML few-shot learning pipeline with a ResNet50 backbone for tree species classification at Danongdafu Forest Park, achieving **70.72% accuracy** with limited labeled data.
- Achieved **92.59% carbon sink estimation accuracy** against a ground-truth survey; estimated monitoring cost savings exceeded USD 167K over 30 years.
- Preprocessed UAV GeoTIFF imagery, annotated tree crowns, and validated geospatial data using QGIS and canopy height model (CHM) analysis.

Advisor: Prof. Wei-Che Chien **Team member:** Aaron Kuo

Publication & Awards

Carbon Sink Estimation in Primary Forests via Individualized Tree Segmentation and Species-Level Classification. Supharada Jundok and Wei-Che Chien. 7th International Conference on Big-data Service and Intelligent Computation (BDSIC 2025), Bangkok, Thailand, Oct 2025.

Best Presentation Award, BDSIC 2025 · **Honorable Mention**, 2025 NDHU CSIE Undergraduate Project Exhibition.

Projects

Alzheimer's Disease Prediction Using Machine Learning — Dec 2025
Big Data Analytics – National Dong Hwa University — *Project*

- Evaluated 11 classifiers; Random Forest achieved **96.74% test accuracy** and **95.39% F1-score**; used 5-fold cross-validation for model evaluation.
- Built an EDA-driven feature selection pipeline using chi-square tests, t-tests, Pearson/Spearman correlations, and heatmaps, reducing **35 features to 5**.
- Addressed class imbalance with `scale_pos_weight` and extended the analysis with SHAP for interpretability.

Scores: Presentation 95/100; Paper 100/100. **Advisor:** Prof. Chung Yung; TA: Demiah Kisla Charlery.

Certifications & Languages

Google Cybersecurity Professional Certificate – Coursera, Google × NTUT — Jul – Dec 2024
Eight certificates covering network security, SIEM, IDS, threat analysis, and security automation with Python.

Languages: Thai & Laotian (Native); English (Professional, TOEIC 825); Traditional Chinese (Basic, TOCFL A2); Spanish (Basic).